Effects of increased dietary protein-to-carbohydrate ratios in women with polycystic ovary syndrome.

BACKGROUND: Some evidence has suggested that a diet with a higher ratio of protein to carbohydrates has metabolic advantages in the treatment of polycystic ovary syndrome (PCOS). OBJECTIVE: The objective of this study was to compare the effect of a high-protein (HP) diet to a standard-protein (SP) diet in women with PCOS. DESIGN: A controlled, 6-mo trial was conducted in 57 PCOS women. The women were assigned through rank minimization to one of the following 2 diets without caloric restriction: an HP diet (>40% of energy from protein and 30% of energy from fat) or an SP diet (<15% of energy from protein and 30% of energy from fat). The women received monthly dietary counseling. At baseline and 3 and 6 mo, anthropometric measurements were performed, and blood samples were collected. RESULTS: Seven women dropped out because of pregnancy, 23 women dropped out because of other reasons, and 27 women completed the study. The HP diet produced a greater weight loss (mean: 4.4 kg; 95% CI: 0.3, 8.6 kg) and body fat loss (mean: 4.3 kg; 95% CI: 0.9, 7.6 kg) than the SP diet after 6 mo. Waist circumference was reduced more by the HP diet than by the SP diet. The HP diet produced greater decreases in glucose than did the SP diet, which persisted after adjustment for weight changes. There were no differences in testosterone, sex hormone-binding globulin, and blood lipids between the groups after 6 mo. However, adjustment for weight changes led to significantly lower testosterone concentrations in the SP-diet group than in the HP-diet group. CONCLUSION: Replacement of carbohydrates with protein in ad libitum diets improves weight loss and improves glucose metabolism by an effect that seems to be independent of the weight loss and, thus, seems to offer an improved dietary treatment of PCOS women.